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Software Project Management

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CSE

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Chapter-4
Software Project Management

- Software quality and management

Software Quality Management

- Software Quality shows how good and reliable a product is.
- Software Quality Management is a process that ensures the required level of software quality is achieved when it reaches the users, so that they are satisfied by its performance.
- The process involves quality assurance, quality planning, and quality control. This tutorial provides a complete overview of Software Quality Management and describes the various steps involved in the process.

Software Quality Management

- Quality software refers to a software which is reasonably bug or defect free, is delivered in time and within the specified budget, meets the requirements and/or expectations, and is maintainable. In the software engineering context, software quality reflects both **functional quality** as well as **structural quality**.
- **Software Functional Quality** − It reflects how well it satisfies a given design, based on the functional requirements or specifications.
- **Software Structural Quality** − It deals with the handling of non-functional requirements that support the delivery of the functional requirements, such as robustness or maintainability, and the degree to which the software was produced correctly.

Software Quality Management

- **Software Quality Assurance** – Software Quality Assurance (SQA) is a set of activities to ensure the quality in software engineering processes that ultimately result in quality software products. The activities establish and evaluate the processes that produce products. It involves process-focused action.
- **Software Quality Control** – Software Quality Control (SQC) is a set of activities to ensure the quality in software products. These activities focus on determining the defects in the actual products produced. It involves product-focused action.
- **QA focuses on preventing defects, while QC focuses on identifying and fixing defects. QA activities occur throughout the development process, while QC activities occur after development.**

- The various factors, which influence the software quality, are termed as software factors.

They can be broadly divided into two categories.

- The first category of the factors is of those that can be measured directly such as the number of logical errors, and the second category clubs those factors which can be measured only indirectly.
- For example, maintainability but each of the factors is to be measured to check for the content and the quality control.

McCall's Factor Model

This model (1977) classifies all software requirements into 11 software quality factors. The 11 factors are grouped into three categories – product operation, product revision, and product transition factors.

- **Product operation factors** – Correctness, Reliability, Efficiency, Integrity, Usability.
- **Product revision factors** – Maintainability, Flexibility, Testability.
- **Product transition factors** – Portability, Reusability, Interoperability.

McCall’s Factor Model

1. <u>Product Operation Software Quality Factors</u>

According to McCall’s model, product operation category includes five software quality factors, which deal with the requirements that directly affect the daily operation of the software. They are as follows −

Correctness These requirements deal with the correctness of the output of the software system.

- Output Mission
- The required accuracy of output that can be negatively affected by inaccurate data or inaccurate calculations.
- The completeness of the output information, which can be affected by incomplete data.
- The up-to-dateness of the information defined as the time between the event and the response by the software system.
- The availability of the information.
- The standards for coding and documenting the software system.

McCall's Factor Model

Reliability

- Reliability requirements deal with service failure. They determine the maximum allowed failure rate of the software system, and can refer to the entire system or to one or more of its separate functions.

Efficiency

- It deals with the hardware resources needed to perform the different functions of the software system. It includes processing capabilities (given in MHz), its storage capacity (given in MB or GB) and the data communication capability (given in MBPS or GBPS).
- It also deals with the time between recharging of the system's portable units, such as, information system units located in portable computers, or meteorological units placed outdoors.

Integrity

- This factor deals with the software system security, that is, to prevent access to unauthorized persons, also to distinguish between the group of people to be given read as well as write permit.

Usability

- Usability requirements deal with the staff resources needed to train a new employee and to operate the software system.

2. Product Revision Quality Factors

According to McCall’s model, three software quality factors are included in the product revision category. These factors are as follows −

- Maintainability

This factor considers the efforts that will be needed by users and maintenance personnel to identify the reasons for software failures, to correct the failures, and to verify the success of the corrections.

- Flexibility

This factor deals with the capabilities and efforts required to support adaptive maintenance activities of the software. These include adapting the current software to additional circumstances and customers without changing the software. This factor’s requirements also support perfective maintenance activities, such as changes and additions to the software in order to improve its service and to adapt it to changes in the firm’s technical or commercial environment.

McCall’s Factor Model

Testability

- Testability requirements deal with the testing of the software system as well as with its operation. It includes predefined intermediate results, log files, and also the automatic diagnostics performed by the software system prior to starting the system, to find out whether all components of the system are in working order and to obtain a report about the detected faults. Another type of these requirements deals with automatic diagnostic checks applied by the maintenance technicians to detect the causes of software failures.

3.Product Transition Software Quality Factor

According to McCall’s model, three software quality factors are included in the product transition category that deals with the adaptation of software to other environments and its interaction with other software systems. These factors are as follows −

Portability

- Portability requirements tend to the adaptation of a software system to other environments consisting of different hardware, different operating systems, and so forth. The software should be possible to continue using the same basic software in diverse situations.

McCall’s Factor Model

Reusability

- This factor deals with the use of software modules originally designed for one project in a new software project currently being developed. They may also enable future projects to make use of a given module or a group of modules of the currently developed software. The reuse of software is expected to save development resources, shorten the development period, and provide higher quality modules.

Interoperability

- Interoperability requirements focus on creating interfaces with other software systems or with other equipment firmware. For example, the firmware of the production machinery and testing equipment interfaces with the production control software.

Interoperability refers to the degree to which a software system, devices, applications or other entity can connect and communicate with other entities in a coordinated manner without effort from the end user.

Responsibilities of management in Software Quality

Basically, a three-level structure of management exists in software development organizations –

- Top management
- Department management
- Project management

Responsibilities of management in Software Quality

Top Management Responsibilities in Software Quality

Following are the responsibilities of the top management in ensuring Software Quality −

- Assure the quality of the company's software products and software maintenance services
- Communicate the importance of the product and service quality in addition to customer satisfaction to employees at all levels
- Assure satisfactory functioning and full compliance with customer requirements
- Ensure that quality objectives are established for the organization's SQA system
- Initiate planning and oversee implementation of changes necessary to adapt the SQA system to major internal as well as external changes related to the organization's clientele, competition, and technology
- Intervene directly to support resolution of crisis situations and minimize damages

Responsibilities of management in Software Quality

Department Management Responsibilities for SQA

- Middle management's quality assurance responsibilities include –
- Management of the software quality management system (quality system-related tasks)
- Management of tasks related to the projects and services performed by units or teams under the specific manager's authority (project-related tasks)

Responsibilities of management in Software Quality

<u>Project management responsibilities on software quality</u>

- Most project management responsibilities are defined in procedures and work instructions; the project manager is the person in-charge of making sure that all the team members comply with the said procedures and instructions.
- His tasks include professional hands-on and managerial tasks, particularly the following −
- **Professional hands-on tasks**
  - Preparation of project and quality plans and their updates
  - Participation in joint customer–supplier committee
  - Close follow-up of project team staffing, including attending to recruitment, training and instruction

Responsibilities of management in Software Quality

- Management tasks
- Project managers address the follow-up issues such as −
  - Performance of review activities and the consequent corrections
  - Software development and maintenance unit’s performance, integration and system test activities as well as corrections and regression tests
  - Performance of acceptance tests
  - Software installation in remote customer sites and the execution of the software system by the customer
  - SQA training and instruction of project team members
  - Schedules and resources allocated to project activities
  - Customer requests and satisfaction
  - Evolving project development risks, application of solutions and control of results

References

- https://www.tutorialspoint.com/software_quality_management/software_quality_management_role_of_management_in_qa.htm
- https://www.tutorialspoint.com/software_quality_management/software_quality_management_factors.htm
- https://www.tutorialspoint.com/software_quality_management/software_quality_management_factors.htm

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